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Started on Monday, 27 February 2023, 6:11 PM

State Finished

Completed on Monday, 27 February 2023, 7:41 PM

Time taken 1 hour 30 mins

Grade 79.83 out of 100.00

Question 1

Partially correct

Mark 8.00 out of 10.00

Consider the following binary classification problem in $\mathbb{R}^2$:

The points (x_1, x_2) labeled with +1 are: $\{(1, 1), (1, 2), (0, 2), (0, 1)\}$

The points (x_1, x_2) labeled with -1 are: $\{(5, 5), (4, 4), (6, 4), (6, 7)\}$

What is the area of the convex hull of the points labelled with +1?

Answer: ✓

What is the area of the convex hull of the points labelled with -1?

Answer: ✓

What is the support vector of the positive class?

Answer: ✓

What is the support vector of the negative class?

Answer: ✓

Compute the margin of the linear SVM classifier that separates the two classes.

margin = ✗

IMPORTANT: For answers with an option, choose an option if you are certain that it is correct. If you choose no option, no points will be deducted. If you choose a wrong option, it will cost you as many points as you would have gotten with the right answer.

Question 2

Not answered

Not graded

This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

Question 3

Correct

Mark 10.00 out of 10.00

Consider the following dataset which is already split into three folds:

	F1		F2		F3	
x	-1	0	0	1	-3	2
y	1	-1	-1	1	-1	1

Assume that you want to train a threshold model $g(x, w)$ with parameter w given by:

$$g(x, w) = \begin{cases} +1 & \text{if } x \geq w \\ -1 & \text{else} \end{cases}$$

Given a training set $\{(x_i, y_i)\}_{i=1}^l$, w is determined by the average sum of absolute error on these data,

$$\text{i.e. } w = \frac{1}{l} \sum_{i=1}^l |x_i - y_i|$$

In order to estimate the generalization error with cross validation, solve the following tasks.

- What is the resulting parameter w , if the first fold is considered for evaluation? Answer: ✓
- What is the resulting risk for the 1st fold, i.e. $L_{cv,1}$, if the 0-1-loss is used as loss function? Answer: ✓

You can use a pocket calculator. Give the answer as a real number rounded to two decimal places. Use "." for the decimal point. Number format example: 1.23

Question **4**

Not answered

Not graded

This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

Question 5

Correct

Mark 10.00 out of 10.00

Convolution:

Consider the input image

$$\mathbf{x} = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 2 & -1 & -1 & 1 & 1 \\ 1 & -1 & 0 & 0 & -1 \\ 2 & 1 & 1 & 2 & 0 \end{pmatrix}$$

and the filter

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

Let $\mathbf{s} = \mathbf{W} * \mathbf{x}$.

Notation: Let $s_{i,j}$ denote the entry of $\mathbf{s}$ in row i and column j . Indices start at 1 (i.e. $s_{1,1}$ denotes the top left entry of $\mathbf{s}$).

The values inserted for the padding operation shall all be zeros!

Assuming stride equal to 1 and padding equal to 2, compute $s_{3,3}$.

Assuming stride equal to 2 and padding equal to 2, the output matrix has d_r rows and d_c columns. Compute d_r .

Assuming stride equal to 1 and padding equal to 1, the output matrix has D_r rows and D_c columns. Compute D_c .

$$s_{3,3} = \boxed{2} \quad \checkmark$$

$$d_r = \boxed{4} \quad \checkmark$$

$$D_c = \boxed{5} \quad \checkmark$$

Question 6

Not answered

Not graded

This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

Question 7

Correct

Mark 10.00 out of 10.00

Consider the following result of a model $\bar{g}$ for a binary classification task. y denotes the true labels.

$\bar{g}(x)$	y
0.32	-1.0
0.53	1.0
-0.16	1.0
-0.19	-1.0
-0.33	-1.0

- What is the Area Under Curve (AUC) of the associated ROC-curve? Answer: ✓
- What are the TRP and FPR of the threshold $\theta = 0.1$ for the associated threshold model $g(x) = \text{sgn}(\bar{g}(x) - \theta)$?
 FPR: ✓
 TPR: ✓

You can use a pocket calculator. Give the answer as a real number between 0 and 1, rounded to two decimal places. Do not give the answer in %. Use "." for the decimal point. Number format example: 0.67

Question 8

Not answered

Not graded

This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

Question 9

Partially correct

Mark 6.00 out of 10.00

Assume we have a possibly unfair coin. For a coin flip there can be two possible outcomes: head or tail. We denote the event of the coin landing on head with 1 and the event of the coin landing on tail with 0.

Let's call the probability of the coin landing on head p . Note that a fair coin would have $p = 0.5$.

To find the parameter p , we conduct an experiment. We flip the coin 4 times and get the following result: $\{1, 1, 1, 0\}$.

a) Following the idea of maximum likelihood which p is a better choice?

An estimator for p that leads to a lower "negative log-likelihood" of the samples is better



For the following you can use a pocket calculator. For the log use the logarithm with base e .

b) Assuming a fair coin, namely $p = 0.5$, what is the negative log-likelihood of getting the result of our experiment?

2.77



c) What is the negative log-likelihood of the received samples if we assume $p = 0.58$?



IMPORTANT: Only choose an answer if you are sure that it is correct. If you choose no answer, no points will be deducted. If you choose a wrong answer, it will cost you as many points as you would have gotten with the right answer.

Question 10

Not answered

Not graded

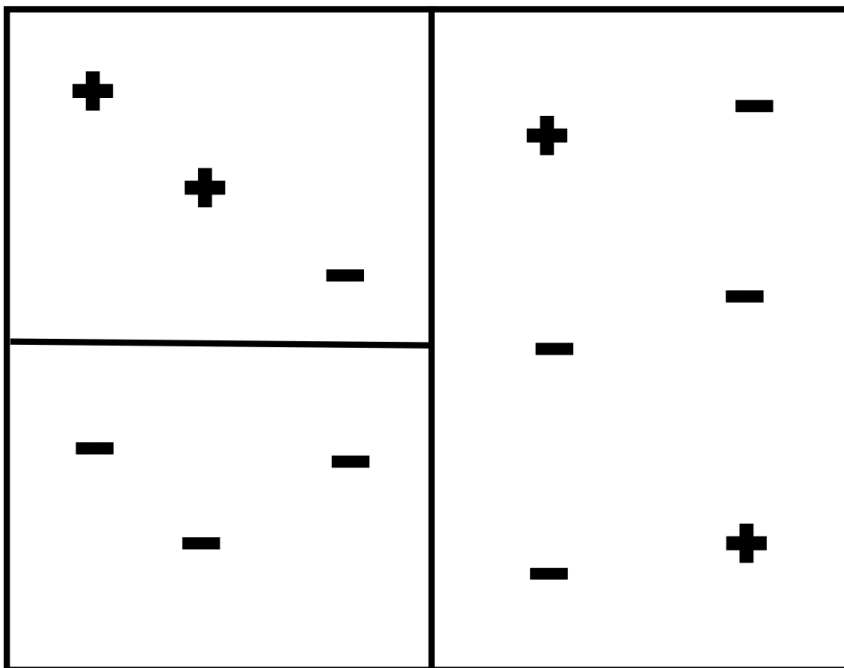
This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

Question 11

Correct

Mark 10.00 out of 10.00

Consider the result of training a decision tree which partitions the depicted binary data (classes: + and -) into three regions.



What is the Gini impurity of the partition on the right side (with 6 datapoints)? ✓

What is the Gini purity gain of the resulting partition? ✓

Question 12

Not answered

Not graded

This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

Question 13

Partially correct

Mark 3.33 out of 10.00

Consider the **general setup for the bias-variance tradeoff** in supervised learning. What expressions are random variables/quantities? (multiple answers possible)

Select one or more:

- ☒ a. The training set $\mathbf{Z}_l$. ✓
- ☐ b. The model class $g(\cdot; \cdot)$.
- ☐ c. The label y for a given $\mathbf{x}_0$.
- ☐ d. The model weights $\mathbf{w}(\mathbf{Z}_l)$ computed by the training set.
- ☐ e. The number of samples l .
- ☐ f. The input $\mathbf{x}_0$.

Question 14

Partially correct

Mark 7.50 out of 10.00

Choose the correct statements involving FNNs, CNNs and RNNs (several may be correct):

Select one or more:

- ☐ a. Sliding a kernel with three channels over an input image of three channels produces an output image that also consists of three channels.
- ☐ b. The weight matrices that process the inputs in an RNN are the same for each sequence.
- ☐ c. In an LSTM block, the current cell state is computed by multiplying together the previous cell state with the current cell input. This procedure prevents the corresponding gradients from vanishing.
- ☒ d. Padding can be used, to control the output dimension of the CNN layer. ✓
- ☒ e. A convolution layer without activation function is linear in terms of the weights. ✓
- ☐ f. The universal function approximator theorem provides a useful algorithm to compute the optimal weights of neural network for approximating any given continuous function.
- ☒ g. Weight-sharing for CNNs means that the same weights are used for different positions of an image. ✓
- ☐ h. RNNs allow to deal with input sequences of variable sequence length and variable number of features.

Question 15

Partially correct

Mark 5.00 out of 10.00

Tick the correct statements related to ensemble methods (several may be correct):

Select one or more:

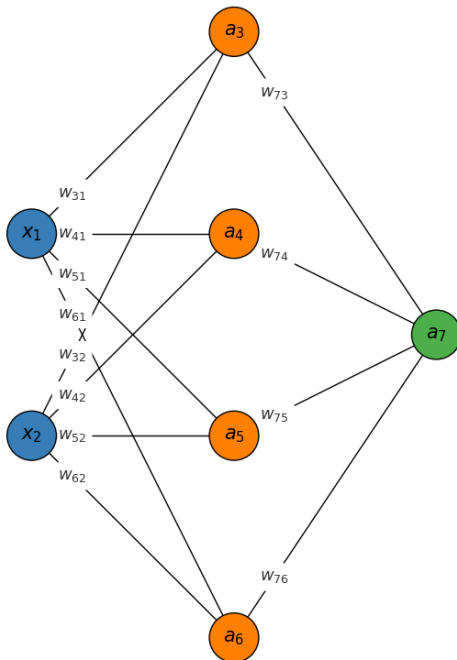
- ☐ a. To construct the out of bag random forest predictor corresponding to a specific data sample $\mathbf{z}$, you only average trees which are trained on data that contain $\mathbf{z}$.
- ☐ b. In AdaBoost, models with larger error rate get assigned the larger model weights.
- ☐ c. In each iteration step, forward stagewise additive modelling tries to compute the optimal weights and basis functions to add to the model from the previous iteration step.
- ☐ d. Each tree in a random forest uses the same subsample of the training data.
- ☐ e. In gradient boosted trees for regression, you fit a new regression tree to the gradient of the loss in each iteration step. For each training datapoint this gradient is evaluated at the output of the model from the previous iteration.
- ☒ f. In each tree of a random forest you randomly select a given number of features in each split. ✓
- ☒ g. Mean accuracy decrease is a method to measure feature importance in random forests. It is based on comparing ✓ out of bag errors before and after permuting a specific feature among the data.
- ☐ h. In each iteration step of AdaBoost, you use the model weights of the previous steps to obtain the current weighted error rate.

Question 16

Correct

Mark 10.00 out of 10.00

Consider the following neural network:



The input unit 1 has activation x . The preactivations of the hidden units $i = 3, 4, 5, 6$ are denoted as s_i and their activations as a_i , respectively.

The preactivation of the output unit 7 is denoted as s_7 and the output as $\hat{y} = a_7$. The true label is denoted as y . There are no biases.

In the hidden layer, we use Relu as activation function, i.e. $f(s) = \text{Relu}(s) = \max(0, s)$.

In the output layer, the activation is the identity function.

The delta-errors at the output is denoted as δ_7 , and the hidden delta-errors as $\delta_3, \delta_4, \delta_5$ and δ_6 respectively.

As loss function, we use the squared error, i.e. $L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$.

The learning rate is denoted as η . The network weights are updated via backpropagation and gradient descent, and these updated weights are denoted by superscript "new".

We are given the input value, the current network weights, the true label, and the learning rate for gradient descent update:

$$x_1 = 1.0, x_2 = 1.0$$

$$w_{31} = -1.0, w_{41} = -1.0, w_{51} = 2.0, w_{61} = -2.0$$

$$w_{32} = 2.0, w_{42} = -2.0, w_{52} = 3.0, w_{62} = -3.0$$

$$w_{73} = 2.0, w_{74} = 1.0, w_{75} = 5.0, w_{76} = -3.0$$

$$y = 19.0$$

$$\eta = 0.05$$

Compute the following quantities (forward pass and backpropagation):

$$\hat{y} = \boxed{27.00} \quad \checkmark$$

$$\frac{\partial L}{\partial w_{73}} = \boxed{8.00} \quad \checkmark$$

$$\frac{\partial L}{\partial w_{31}} = \boxed{16.00} \quad \checkmark$$

$$w_{73}^{\text{new}} = \boxed{1.60} \quad \checkmark$$

Question **17**

Not answered

Not graded

This textfield is optional and will not influence the automatic grading. Here you can write down subresults or your thoughts on the specific tasks. It can be used to your advantage at the end of the semester as mentioned in the exam info file.

◀ Exam 2 - Endterm - (01.02.2023, 16:00)

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